

Prevention and Nondialytic Management of Acute Kidney Injury

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Acute kidney injury (AKI) often results from a combination of insults. The most commonly associated causes are failure of kidney autoregulation, direct nephrotoxicity, ischemia/reperfusion, and inflammatory states. AKI severity predicts adverse outcomes, including requirement for kidney replacement therapy (KRT), length of hospital stay, and mortality. Studies using the Risk, Injury, Failure, Loss of Kidney Function, and End-Stage Kidney Disease (RIFLE); Acute Kidney Injury Network (AKIN); and Kidney Disease Improving Outcomes (KDIGO) classification systems (see [Chapter 72](#)) have shown that even small changes in creatinine levels are associated with short-term and long-term mortality.^{1,2} Furthermore, the distant effects of AKI contribute to dysfunction of other organs, such as the heart, lungs, brain, and liver.³ Consequently, primary prevention and early diagnosis of AKI are of primary importance. Once AKI has been detected, efforts to attenuate the effects of injury and treatment of its consequences are necessary.

RISK ASSESSMENT

Considering the conceptual model of AKI illustrated in [Fig. 74.1](#), the first step in preventing AKI is an adequate risk assessment. The initial care of patients at risk should focus on identification and, if possible, reversal of the risk factors. Several factors are associated with AKI ([Box 74.1](#)). Risk scores are also available to help predict AKI development; most of them focus on specific situations (see [Box 74.2](#)). However, in most clinical settings, there are scarce data to evaluate the potential impact of modifiable risk factors on AKI development and progression.

The most common identifiable risk factor is chronic kidney disease (CKD), and thus adequate determination of baseline kidney function is fundamental. Because 40% to 50% of patients do not have a baseline kidney function available, and the first creatinine measured in the hospital may be affected by the disease process occurring before hospital admission, CKD status is sometimes difficult to determine. In addition to the damage biomarkers, markers of kidney stress such as TIMP-2*IGFBP7 have been shown to be helpful in identifying ongoing injury and enabling early assessment. For a further discussion of risk factors and scoring systems, see [Chapters 72 and 73](#).

PRIMARY PREVENTIVE MEASURES

Optimizing Volume and Hemodynamic Status

To prevent AKI, ensuring adequate kidney perfusion is essential. Prompt resuscitation of patients with hypoperfusion was shown to improve outcomes twenty years ago. Means of optimizing perfusion pressure include administration of fluid and the improvement of hemodynamic status and cardiac output. Common reasons for fluid administration and/or use of vasopressors to prevent AKI include hypovolemia, hypotension, and sepsis. However, assessment of volume

status can be challenging, particularly in patients in the intensive care unit (ICU). In addition, studies in ICU patients or patients undergoing surgery have shown that only about half of hemodynamically unstable patients respond to fluid administration. There are no guidelines for optimizing hemodynamic and fluid status for kidney function preservation. Too often, the effect of fluid expansion on hemodynamic status and kidney function is retrospective and evaluated by trial and error. However, dynamic measures such as the passive leg raising maneuver and the fluid bolus test coupled with real-time stroke volume monitoring can determine fluid responsiveness accurately.^{4,5} Unfortunately, these are rarely performed even in ICUs. On the ward, volume status assessment and response to fluid administration most often relies on blood pressure, heart rate, oxygen saturation, central venous pressure, and urine output. These parameters are not specific or sensitive, and thus better simple tools are urgently needed to assess volume status and fluid responsiveness outside ICUs. In emergency settings, point-of-care ultrasound can be useful. For example, a collapsing inferior vena cava at the end of expiration is suggestive of hypovolemia.

Four phases of fluid therapy have been conceptualized, including rescue, optimization, stabilization, and deescalation.⁶ *Rescue* implies the “administration of fluid for immediate management of life-threatening conditions associated with impaired tissue perfusion.” *Optimization* refers to the “adjustment of fluid type, rate, and amount based upon context to achieve optimization of tissue perfusion.” Fluid boluses are used during the rescue phase, whereas fluid challenges are administered during the optimization phase. Fluid bolus in an adult typically includes the infusion of 500 mL of isotonic fluids over 15 minutes or less without close monitoring. Fluid challenge involves the administration of 250 mL or 3 mL/kg of isotonic fluids over 5 to 10 minutes with stroke volume (SV) reassessment. Some studies have defined fluid responders as an increase of 10% to 15% of SV or cardiac output after fluid challenge. These two phases are essential in preventing AKI secondary to hypoperfusion. *Stabilization* aims for achieving a neutral or slightly negative fluid balance to favor organ support, and *de-escalation* is defined by the “minimization of fluid administration, mobilization of extra fluid to optimize fluid balance.” In the stabilization and de-escalation phases, clinicians should target a neutral and then a negative fluid balance if fluid overload is present. To obtain a neutral or negative fluid balance, fluid administration should be minimized and oral or intravenous (IV) diuretics or even ultrafiltration may be required depending on the clinical scenario and underlying kidney and cardiac function. Adverse outcomes associated with fluid overload include cardiopulmonary complications, delayed wound healing and kidney function recovery, and increased mortality.^{7,8} A meta-analysis has confirmed an association between elevated central venous pressure most often measured within 24 hours of ICU admission and AKI, suggesting a role of venous congestion in the development of AKI.⁹

Conceptual Model for AKI

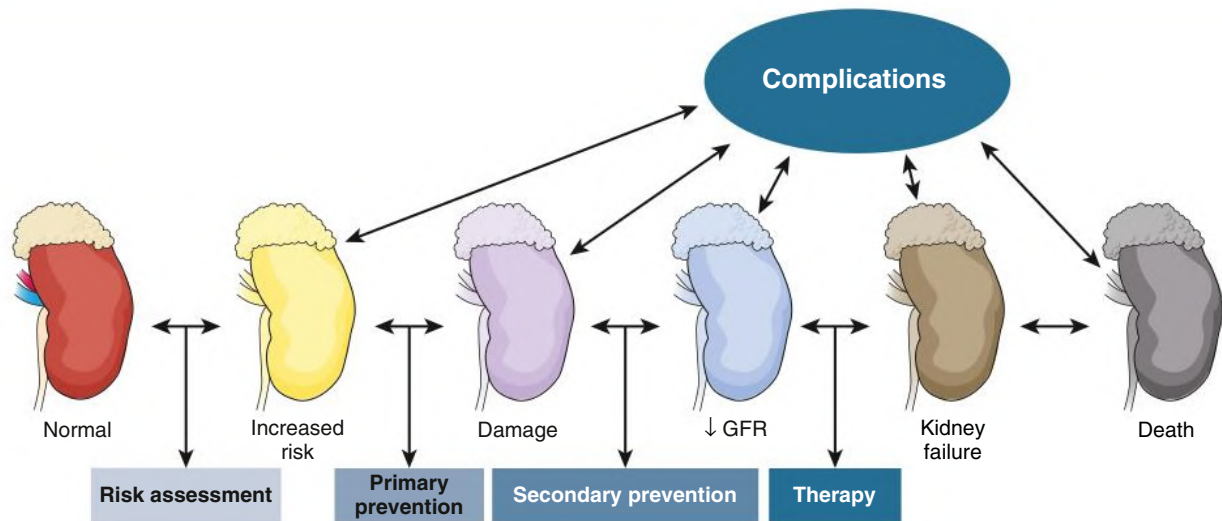


Fig. 74.1 Conceptual model for acute kidney injury (AKI). *GFR*, Glomerular filtration rate. (Modified from Mehta RL, Kellum JA, Shah SV, et al. Acute Kidney Injury Network: Report of an initiative to improve outcomes in acute kidney injury. *Crit Care*. 2007;11:R31.)

BOX 74.1 Major Risk Factors for Acute Kidney Injury

Patient Factors

- Older age (>75 years)
- Diabetes
- Liver failure
- Chronic kidney disease
- Atherosclerosis
- Kidney artery stenosis
- Hypertension
- Hypotension
- Hypercalcemia
- Hyperuricemia
- Sepsis
- Perioperative cardiac dysfunction
- Rhabdomyolysis
- Tumor lysis syndrome

Medications and Agents

- Nonsteroidal antiinflammatory drugs
- Cyclooxygenase-2 inhibitors
- Cyclosporine or tacrolimus
- Angiotensin-converting enzyme inhibitors
- Angiotensin receptor blockers
- Iodinated radiocontrast agent
- Hydroxyethyl starch
- Aminoglycosides
- Amphotericin

Procedures

- Cardiopulmonary bypass procedures
- Surgery involving aortic clamp
- Increased intra-abdominal pressure
- Large arterial catheter placement with risk for atheroembolization
- Liver transplantation
- Kidney transplantation

There is controversy about the optimal fluid to use for resuscitation. The 2012 KDIGO AKI guideline suggests that isotonic crystalloids should be used instead of synthetic (hydroxyethyl starch [HES]) and nonsynthetic (albumin) colloids for intracellular volume expansion in patients at risk or presenting with AKI, in the absence of hemorrhagic shock.¹⁰ Large studies have clearly demonstrated that HES solutions increase the risk for AKI. A randomized controlled trial (RCT) in 7000 ICU patients has shown that even solutions with lower molecular weight such as 6% HES 130/0.4 increased the need for KRT compared with 0.9% sodium chloride (normal saline).¹¹ Another large RCT including 804 patients with severe sepsis has shown that 6% HES 130/0.4 is detrimental to kidney function and also survival compared with Ringer acetate.¹² The mechanism of HES-induced AKI may be because of proximal kidney epithelial cell uptake of HES causing an acquired lysosomal storage disease. Therefore, HES should be avoided in patients at risk for and with AKI.

For albumin, the Saline Versus Albumin Fluid Evaluation (SAFE) trial of 6997 critically ill patients found that fluid resuscitation with saline or albumin resulted in similar relative risks for death.¹³ In addition, no significant differences were found in new single-organ and multiple-organ failure or days on KRT. Two subgroup analyses from this study showed that use of albumin may be deleterious in patients with traumatic brain injury and potentially beneficial in sepsis. Albumin can be considered when substantial amounts of crystalloids (e.g., 2 L) are required to maintain adequate mean arterial pressure (MAP), especially in septic patients. However, its effect must be balanced against its potential risks; that is, it is possibly deleterious in patients with trauma and has low potential for transmission of infectious diseases.

The type of crystalloids used can also influence kidney outcomes. When comparing kidney artery flow velocity and cortical tissue perfusion, there was a significant reduction in artery flow and cortical tissue perfusion with isotonic saline but not with the use of chloride-restrictive fluids. Indeed, greater chloride delivery to the macula densa may activate tubuloglomerular feedback, triggering kidney vasoconstriction and reduced glomerular filtration rate (GFR). Two large single-center RCTs have shown beneficial effects of chloride-restrictive

BOX 74.2 Specific Risk Factors for the Development of Acute Kidney Injury in Common Clinical Situations

Cardiac Surgery

Female sex
COPD
Proteinuria
Preoperative $S_{Cr} > 2.1$ mg/dL
Anemia
Insulin-dependent diabetes
ACE inhibitor therapy
Heart failure
LV ejection fraction $< 35\%$
Preoperative IABP
Hyperglycemia
Hyperuricemia
Emergency surgery
Valve Surgery only
Previous cardiac surgery
Other cardiac surgery
Combination of CABG + valve surgery
Blood transfusion

Critically Ill

High A–a gradient
Low serum albumin
Proteinuria
Hyperglycemia
High intra-abdominal pressure
Active cancer
Age
 $SCr > 1.3$ mg/dL
Serum bilirubin > 1.5 mg/dL
Elevated CVP > 8 cm
Hemodynamic instability
Sepsis

Gastrointestinal and Endocrine

Cirrhosis/biliary surgery
Obstructive jaundice
Diabetes mellitus
Hyperglycemia

Hemodynamic

Congestive heart failure
Aortic cross-clamping
Cardiac instability
Major vascular surgery
Infection/sepsis
Multiorgan failure

Iodinated Contrast Administration

Age > 75 yr
Heart failure
Diabetes mellitus
 $SCr > 1.5$ mg/dL or $eGFR < 60$ mL/min/1.73 m²
History of pulmonary edema
Anemia/blood loss ($Hct < 395$ for men, $< 36\%$ for women)
Hyperuricemia
Volume of contrast > 100 mL
Intra-arterial injection
Systolic BP < 80 mm Hg for > 1 hour and need for inotropic support or IABP 24 hours after procedure
Use of IABP

Nephrotoxic Antibodies

Volume depletion
Older age
Chronic kidney disease
Amphotericin
Aminoglycosides
Duration of therapy > 7 days
Volume depletion
Divided dose regimens
Liver disease

Miscellaneous

Age
Proteinuria
Hypertension
Massive blood transfusion

^aA–a gradient is the alveolar-arterial oxygen gradient calculated with the sea level standard formula $(713 \times FiO_2) - (P_{CO_2}/0.8) - PaO_2$, where FiO_2 is fractional inspired oxygen concentration, PaO_2 is arterial partial oxygen pressure, and P_{CO_2} is partial carbon dioxide pressure.

ACE, Angiotensin-converting enzyme; BP, blood pressure; CABG, coronary arterial bypass graft; COPD, chronic obstructive pulmonary disease; CVP, central venous pressure; Hct , hematocrit; IABP, intra-aortic balloon pump; LV, left ventricular; SCr , serum creatinine.

fluids.^{14,15} In critically ill adults, chloride-restrictive fluids (lactated Ringer's solution or Plasma-Lyte) compared with isotonic saline decreased major kidney events (MAKE), defined as the composite outcome of death from any cause, new kidney replacement therapy, or persistent kidney dysfunction (14.3% vs. 15.4%; odds ratio [OR], 0.91; 95% confidence interval [CI], 0.84–0.99; $P = .04$).¹⁴ In noncritically ill adults, chloride-restrictive fluids (lactated Ringer's solution or Plasma-Lyte) also decreased the incidence of MAKE compared with isotonic saline (4.7% vs. 5.6%; OR, 0.82; 95% CI, 0.70–0.95; $P = .01$).¹⁵ A meta-analysis in critically ill patients has found that resuscitation with chloride-restrictive fluids decreased 30-day mortality and need for KRT compared with isotonic saline.¹⁶ Other RCTs are underway to confirm these findings (PLUS and BASICS trials). The choice of chloride-rich (isotonic saline) versus low-chloride solutions (lactated

Ringer's solution or Plasma-Lyte) should be based on the patient's clinical condition, including electrolyte and acid-base balance. Physicians must also remember that the sodium content of isotonic saline is 154 mEq/L, whereas it is 130 mEq/L for lactated Ringer's and 140 mEq/L for Plasma-Lyte. Of note, normal saline should be used in head-injured patients and patients with severe liver disease or other conditions imposing risk of worsening cerebral edema.

There is no consensus on how to use vasopressors to prevent AKI or avoid further deterioration of kidney function. According to international guidelines for sepsis management, after initial fluid resuscitation with 30 mL/kg of crystalloids and addition of albumin in patients requiring substantial amounts of crystalloids to maintain adequate MAP, vasopressors should be initiated targeting a MAP above 65 mm Hg.¹⁷ In sepsis, norepinephrine is the first-choice

vasopressor. In addition, angiotensin II and vasopressin are noncatecholamine options that may improve glomerular flow through several mechanisms. With regard to the kidney, one RCT showed that vasopressin is equivalent or may be better than norepinephrine to improve kidney outcomes in patients with septic shock.¹⁸ KRT was required in 25% of patients in the vasopressin group and in 35% of the norepinephrine group; however, rates of KRT were comparable among survivors from each group, and the overall mortality rate was also similar between groups.

Inotropic agents such as dobutamine should be administered if myocardial dysfunction or ongoing signs of hypoperfusion are present. The optimal MAP values to prevent AKI need to be better delineated. One study has found that in hypertensive patients with septic shock, targeting MAP between 80 to 85 mm Hg instead of 65 to 70 mm Hg reduced the risk for AKI and KRT, whereas the risk for atrial fibrillation seemed to be heightened.¹⁹ Therefore, target MAP values may need to be individualized depending on age, hypertension status, and degree of peripheral artery and renovascular disease and are likely to be between 60 and 80 mm Hg.

Prevention of Contrast-Induced Acute Kidney Injury

The concept of contrast-induced AKI (CI-AKI) has been recently questioned. However, physicians are less likely to prescribe contrast to patients with lower estimated GFR (eGFR) and more likely to prescribe IV fluids when contrast is administered, which may affect results from observational studies.

According to the KDIGO guideline, measures to reduce the risk of AKI should be implemented in patients with a baseline eGFR below 45 mL/min/1.73 m² for IV iodinated contrast administration. For prevention of CI-AKI, patients at risk should receive IV hydration (Fig. 74.2). Hydration with isotonic saline is superior to half-isotonic (0.45%) saline. When comparing the effectiveness of sodium bicarbonate with isotonic sodium chloride and oral N-acetylcysteine (NAC) with placebo in 8680 high-risk patients scheduled to undergo coronary or noncoronary angiography, the

PRESERVE trial showed no difference in terms of AKI or mortality between the use of sodium bicarbonate and isotonic sodium chloride.²⁰ Therefore, because of the risks of compounding errors with sodium bicarbonate, isotonic sodium chloride should be used. The use of NAC is discussed later in this chapter. Ringer's lactate is another option because it contains less chloride than isotonic sodium chloride, possibly decreasing kidney vasoconstriction and causing less hyperchloremic acidosis. In addition, it causes alkalization of renal tubular fluid, which is a suggested mechanism to decrease the damages caused by free radicals. Studies are needed to determine whether the use of balanced solutions compared with normal saline would lower AKI incidence.

Iodinated contrast can be categorized according to osmolality into high-osmolar contrast medium (approximately 2000 mOsm/kg), low-osmolar contrast medium (600–800 mOsm/kg), and iso-osmolar contrast medium (290 mOsm/kg). The KDIGO AKI guidelines recommend either iso-osmolar (preferred) or low-osmolar iodinated contrast for patients at risk for CI-AKI because the risk for nephrotoxicity is higher with increasing osmolality.

The volume of contrast administered is also an independent predictor of CI-AKI and should be reduced as much as possible. Based on the volume (V) of contrast given (in mL) and the creatinine clearance (CrCl; mL/min), a V:CrCl ratio above 3.7 independently predicts CI-AKI. Administration of iodinated contrast more than once over 48 to 72 hours should be avoided. The drugs used for CI-AKI prevention are included in the section on pharmacologic approaches.

Prevention of Drug-Induced and Nephrotoxin-Induced Acute Kidney Injury

Drug-induced nephrotoxicity often can be predicted because it is more common in certain patient populations and clinical situations. Prevention involves the knowledge of mechanisms of kidney injury, patient-related risk factors, and drug-related risk factors. The most important patient-related factors are older age (60 years or older), CKD, diabetes, heart failure, volume depletion, and sepsis. Preventive

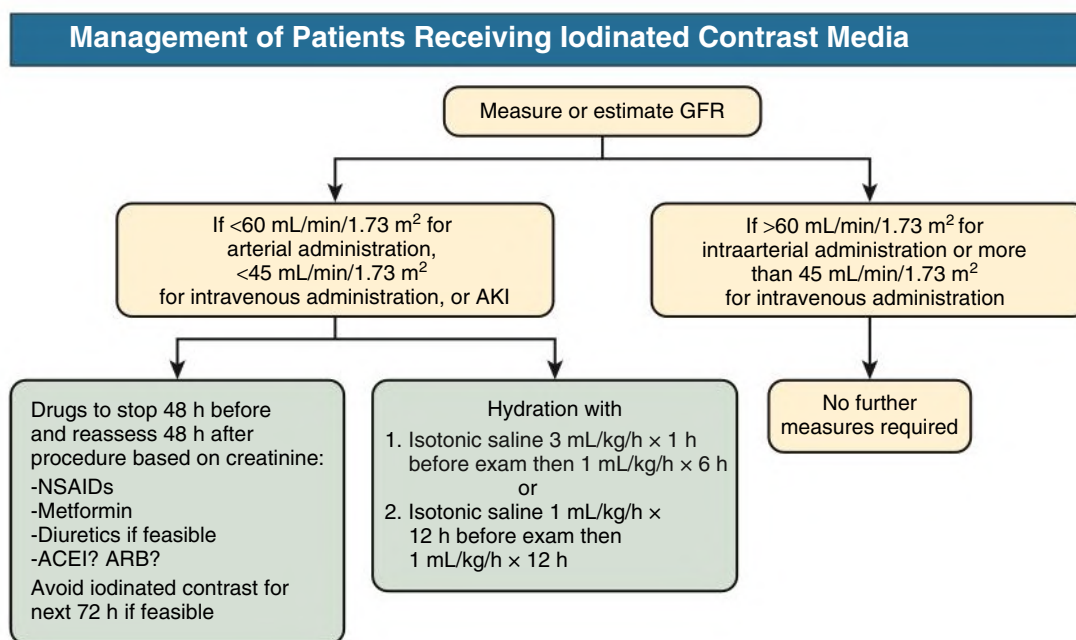


Fig. 74.2 Management of patients receiving iodinated contrast media. ACEI, Angiotensin-converting enzyme inhibitor; AKI, acute kidney injury; ARB, angiotensin receptor blocker; CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration; GFR, glomerular filtration rate; MDRD, Modification of Diet in Renal Disease; NSAIDs, nonsteroidal antiinflammatory drugs.

measures include correctly estimating the GFR before initiating therapy, adjusting the dosage, monitoring kidney function and drug dosage during therapy, and administering administration IV saline before exposure if possible. Alternative nonnephrotoxic drugs should be used, and nephrotoxic drug combinations should be avoided whenever feasible.

Amphotericin

Amphotericin-associated nephrotoxicity can occur in as many as one-third of treated patients, and the risk for AKI increases with higher cumulative doses. Lipid formulations cause less nephrotoxicity compared with the standard formulation, and therefore liposomal amphotericin is preferred over conventional amphotericin; however, it is significantly more expensive. Recently, alternative antifungal agents such as itraconazole, voriconazole, and caspofungin have been more commonly used in patients at high risk for AKI because of their absence of nephrotoxicity and should be used rather than conventional amphotericin when appropriate antifungal coverage is provided.

Angiotensin-Converting Enzyme Inhibitors and Angiotensin Receptor Blockers

Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) cause vasodilation of the efferent glomerular arteriole, further reducing intraglomerular pressure already compromised by the blood pressure-lowering effect of these agents. After the initiation of an ACE inhibitor or ARB, if creatinine increases by more than 30%, bilateral kidney artery stenosis, stenosis of the kidney artery in a solitary kidney, diffuse intrakidney small-vessel disease, or generalized volume depletion should be suspected, and these drugs should be discontinued. It remains unclear whether withdrawing an ACE inhibitor or ARB before iodinated contrast administration is beneficial.

Nonsteroidal Antiinflammatory Drugs

Nonsteroidal antiinflammatory drugs (NSAIDs) should be avoided if possible in CKD and intravascular volume depletion because they inhibit cyclooxygenase, which blocks prostaglandin-induced vasodilation of the afferent arteriole, potentially reducing GFR and kidney blood flow. The risk of NSAIDs must be balanced against their benefit, and patients should be instructed about increased risk situations and advised on when to hold these drugs during sick days. Clinicians may also decide to temporarily stop ACE inhibitors or ARBs in patients with high-normal potassium or more severe CKD. In critically ill patients, kidney hypoperfusion caused by decreased effective circulating volume is relatively common and inhibition of prostaglandin-induced vasodilation may further compromise kidney blood flow and exacerbate ischemic injury.

Aminoglycosides

AKI caused by aminoglycoside nephrotoxicity (AG-AKI) usually occurs 5 to 10 days after initiation of the treatment. AG-AKI is typically nonoliguric and associated with decreased urine concentrating ability and urinary magnesium wasting. Aminoglycoside toxicity can also result in vestibular toxicity that can lead to tinnitus or difficulty ambulating. A large population-based study in adults undergoing surgery receiving antibiotic prophylaxis found an association between one dose of gentamicin (4 mg/kg) and AKI that challenges the notion that AG-AKI only occurs after a few days of treatment.²¹ The KDIGO guideline recommends against using aminoglycosides in patients with or at risk for AKI unless no other alternative is available. With multiple daily administration schedules, elevated aminoglycoside peak levels appear to correlate with nephrotoxicity. Because aminoglycoside uptake by proximal tubular cells is saturable, once-daily administration

can decrease tubular cell toxicity. In the general population, extended intervals between doses maintains the target dose while decreasing the risk for nephrotoxicity compared with multiple daily doses; therefore, in patients with normal eGFR who are not at risk for AKI, aminoglycosides should be administered daily if possible, with serum concentrations measured as recommended.

Tumor Lysis Syndrome

Tumor lysis syndrome (TLS) is caused by uric acid and calcium-phosphate precipitation in the tubules. Identification of patients at high risk is fundamental to prevent AKI in this setting. The most common hematologic malignancies associated with TLS are aggressive lymphomas and acute lymphoblastic leukemia. Rare cases of a TLS-type of AKI have also been reported in other catabolic states, such as after heart surgery. The risk factors for TLS are related to patient (age, baseline kidney function) and tumor characteristics (cell turnover rate, growth rate, extensive bone marrow involvement, tumor bulk, and chemosensitivity).²² Baseline uric acid higher than 7.5 mg/dL, lactate dehydrogenase levels higher than 1500 U/L, and white blood cells greater than $25 \times 10^9/L$ are also risk factors. The diagnosis is based on two simultaneous laboratory abnormalities occurring within 3 days before or 7 days after chemotherapy: uric acid greater than 8 mg/dL, potassium greater than 6 mEq/L, phosphate greater than 4.5 mg/dL, calcium less than 7 mg/dL, and at least one clinical complication among AKI defined as an increase in baseline creatinine by 1.5 times or greater, cardiac arrhythmia, or seizure.

In patients at low risk for developing TLS as defined in an expert consensus, management includes hydration and close monitoring of volume status and kidney function.²² The use of urine alkalinization to promote elimination of urate is not recommended because it can induce calcium phosphate deposition and therefore aggravate TLS. In patients at intermediate and high risk, aggressive hydration with isotonic saline 2 to 3 L/m² per day aiming for urine output between 80 and 100 mL/m²/hr should be initiated. Hydration should be continued until there is no significant evidence of tumor lysis, as shown by uric acid and phosphorus levels. If urine output decreases despite adequate fluid administration, a loop diuretic should be added, and KRT will be required if oliguria persists. In patients at intermediate risk with uric acid levels up to 8 mg/dL, a xanthine oxidase inhibitor such as allopurinol also should be started 2 days before chemotherapy, whereas rasburicase should be used in patients with uric acid levels greater than 8 mg/dL or with concomitant cardiac or kidney diseases limiting hydration. The risk of severe cutaneous adverse events associated with allopurinol appears higher in some Asian populations. Rasburicase should not be used in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency. For high-risk patients, a single dose of rasburicase (up to 0.2 mg/kg, although a lower dose such as 3 mg is usually prescribed) is recommended, followed by close monitoring of uric acid levels. If uric acid normalizes, allopurinol treatment can be started.

Secondary Prevention

After the kidney insult has occurred, secondary preventive measures should be directed to avoid further injury, facilitate repair and recovery, and prevent complications. The timeliness of interventions is crucial to their effectiveness. Various approaches have been applied but are best appreciated in the context of specific clinical scenarios.

Traumatic and Nontraumatic Rhabdomyolysis

The main step in preventing AKI after traumatic rhabdomyolysis and crush syndrome is early and aggressive fluid therapy.²³ IV isotonic saline should be initiated even before the crushed limb is

relieved to prevent precipitation of the pigment in the tubular lumen. Nontraumatic rhabdomyolysis can be caused by ischemia, bacterial or viral infections, severe heat, prescribed medications (statins), drugs of abuse (cocaine or heroin), inflammatory or metabolic myopathies, and electrolyte disorders (hypokalemia or hypophosphatemia). Plasma creatine kinase (CK) levels greater than 5000 U/L have been associated with AKI. In patients with CK levels above 10,000 U/L or with severe crush injury, aggressive IV fluid administration should be considered to correct hypovolemia and target urine output of 200 to 300 mL/h to reduce kidney injury. In patients with less severe rhabdomyolysis with no rapid increase in CK levels (i.e., peak 5000–10,000 U/L) and in those with underlying CKD or heart failure, less aggressive fluid infusion can be targeted. Fluid composition is controversial because no direct comparative trials have been performed. Recommendations to use sodium bicarbonate intend to maintain alkaline urine and decrease precipitation of myoglobin and arteriolar vasoconstriction. However, sodium bicarbonate also can precipitate calcium phosphate deposition and worsen hypocalcemia and should be avoided in severe hypocalcemia or metabolic alkalosis. On the other hand, large volumes of normal saline can cause hyperchloremic metabolic acidosis. Some experts use both normal saline and sodium bicarbonate. For example, if urine pH is less than 6.5, each liter of normal saline can be alternated with 850 mL of 5% dextrose plus 150 mmol of sodium bicarbonate. Bicarbonate should not be administered if there is concomitant hypocalcemia or metabolic alkalosis. Mannitol has been suggested to be beneficial because of its diuretic, antioxidant, and vasodilatory properties. Mannitol could prevent kidney tubular cast deposition, expand extracellular volume, and reduce intracompartmental pressure, muscle edema, and pain. However, mannitol may exacerbate heart failure and nephrotoxicity, requires close monitoring, and is contraindicated in oliguria, hypervolemia, hypertension, and heart failure. Mannitol may be considered in patients with CK levels greater than 30,000 U/L who have a urine flow greater than 20 mL/hr at a rate of 5 g/hr added to each liter of infusate not to exceed 1 to 2 g/kg/day.

Muscle damage induces stretch-activated ion channels, allowing for influx of calcium into cells after reperfusion. The resulting hypocalcemia is usually asymptomatic but can lead to cardiac dysrhythmias. Hence, care must be taken to avoid sodium bicarbonate–induced hypocalcemia, which can trigger tetany, seizures, and cardiotoxicity and worsen muscle damage. During AKI recovery, hypercalcemia is frequent, mainly in patients who received calcium infusion, as a result of the mobilization of previously precipitated calcium. Thus, hypocalcemia should be treated only if symptomatic.

In treating patients with rhabdomyolysis, it is important to consider when to stop fluid resuscitation. A general recommendation is to stop when CK levels decrease to less than 5000 U/L and myoglobinuria disappears, as shown by a negative urine dipstick for blood. However, the risk for fluid accumulation and compartmental expansion always should be evaluated. Frequent assessment of serum creatinine and CK also help the clinician determine the appropriate volume expansion. KRT should be considered in resistant hyperkalemia or metabolic acidosis, rapidly rising serum potassium, oliguria, anuria, or volume overload.

Hyperglycemia

It remains uncertain whether strict control of blood glucose reduces AKI incidence and mortality. In a large RCT in ICU patients, intensive glucose control (glucose of 81–108 mg/dL [4.5–6.0 mmol/L]) increased the risk for death at 90 days compared with conventional glucose control (<180 mg/dL [<10 mmol/L]). Intensive glucose control also increased the risk for severe hypoglycemia. There was no

change in the incidence of AKI or use of KRT. Other studies have not found an increase in mortality with intensive glucose control. In summary, intensive glucose control in ICU patients increased the incidence of severe hypoglycemia and either increased or had no effect on mortality compared with blood glucose ranges of 140 to 180 mg/dL (7.8–10 mmol/L) and 180 to 200 mg/dL (10–11 mmol/L). We recommend maintaining glucose concentration in the range of 110 to 149 mg/dL (6.1–8.3 mmol/L).

Remote Ischemic Preconditioning

Remote ischemic preconditioning (RIPC) is performed by applying inflation of a blood pressure cuff for four or five short cycles in the upper or lower limb. RIPC aims to create brief ischemia and reperfusion in the arm or leg to provide protection in distant organs, such as heart, kidney, lung, and brain. The underlying mechanisms include activation of humoral factors, including adenosine, bradykinin, and cannabinoids, in addition to subcellular modulators, nuclear factor- κ B (NF- κ B), and nitric oxide (NO). A meta-analysis including 28 studies, which randomized a total of 6851 patients, concluded that RIPC does not confer a benefit over sham conditioning (placebo) to prevent AKI or reduce mortality in cardiac or vascular surgery.²⁴

Pharmacologic Approaches

Because of the multiple different causes of AKI, various pathways have been targeted in studies to prevent or alter the course of AKI. These pathways include inflammation, oxidative and mitochondrial stress, cellular metabolism and repair, apoptosis, and hemodynamics. Most of these preventive strategies were successful in animal models but did not translate into beneficial effects in patients. Only a few have shown benefits (Table 74.1).

N-Acetylcysteine

N-Acetylcysteine (NAC) is a tripeptide analogous to glutathione and is able to cross cellular membranes. NAC may reduce vasoconstriction and oxygen free radical generation after the administration of contrast material. Because an increased production of free radicals by the kidneys is partly responsible for their cellular damage in postischemic and nephrotoxic AKI, several clinical studies have attempted to use NAC to prevent AKI, mainly in CI-AKI and during cardiac surgery.

In the initial study, NAC at a dose of 600 mg orally twice daily the day before and the day of the procedure prevented AKI after iodinated contrast administration. The multicenter PRESERVE trial has shown that the use of NAC does not provide any benefit to prevent CI-AKI.²⁰ Therefore, NAC is no longer recommended to prevent CI-AKI.

Allopurinol

Hyperuricemia has been associated with the development of AKI. There are retrospective observational studies showing a reduced incidence of AKI in patients treated with allopurinol compared with those who were not. Animal models of heat stress and rhabdomyolysis with hyperuricemia have also demonstrated that allopurinol administration could prevent AKI. Allopurinol may also help in reducing AKI after iodinated contrast administration. Additional studies are required.

Loop Diuretics and Natriuretics

Diuretics are often used to manage fluid in patients who develop AKI. Although nonoliguric AKI has been associated with better outcomes than oliguric AKI, diuretics have been shown to be ineffective in the prevention of AKI or for improving outcomes once AKI occurs. In addition, diuretics should be avoided when AKI is attributed to pre-renal causes. Meta-analyses have confirmed that the use of diuretics to prevent AKI did not reduce in-hospital mortality or need for KRT.²⁵

TABLE 74.1 Drugs Used in Prevention of Acute Kidney Injury

Drug	Level of Evidence	Results	Comments
Dopamine	RCTs	No effect on kidney function	
Fenoldopam	Small RCTs One meta-analysis	No effect on kidney function Beneficial effect on kidney function	Further studies required
Norepinephrine vs. vasopressin	RCT in septic shock	Tendency toward less kidney replacement therapy with vasopressin	Further studies required
Loop diuretics	RCTs and meta-analysis	No effect on kidney function	
Nesiritide	RCTs in cardiac surgery	Decreased incidence of AKI Controversial effect on kidney replacement therapy requirement	Further studies required
<i>N</i> -Acetylcysteine	RCTs and meta-analysis	No effect in CI-AKI	
Allopurinol	Observational studies and small pilot studies	Possible beneficial effect on kidney function	Further studies required
Statins	RCTs in perioperative period (cardiac surgery) RCTs in CI-AKI	No effect on kidney function Possible beneficial effect on kidney function	Patients not receiving statins who need this drug for other indications may receive statins before their angiography
Insulin	Meta-analyses	Controversial effect	KDIGO recommends targeting blood glucose 110–149 mg/dL (6.1–8.3 mmol/L)
Calcium channel blockers	RCT in peritransplant period	No effect on kidney function	Further studies required
Levosimendan	RCT in cardiac surgery	No effect on need for kidney replacement therapy	
Adenosine antagonists	RCTs	Controversial effect on kidney function	Further studies required
Alpha-melanocyte-stimulating hormone analog (ABT-719)	Phase IIb study in cardiac surgery	No effect on kidney function	
Alpha-2 adrenergic agonist (dexmedetomidine)	Meta-analysis in cardiac surgery Small RCT in aortic surgery	Might decrease AKI Decreased incidence of AKI	
Antioxidative enzyme heme oxygenase-1	Small phase III trial	Trend toward lower levels of the urinary AKI biomarkers	Ongoing phase III in kidney transplantation (NCT03646344)
Propofol	RCTs	Controversial results on kidney function	Current RCTs in kidney transplantation (NCT02727296), after nephrectomy (NCT04474600), orthopedic surgery (NCT03336801)
Small interfering RNA targeting p53	Phase II RCT in cardiac surgery	Beneficial effect on kidney function	Phase III trial ongoing in cardiac surgery (NCT03510897)
Peroxisome proliferator-activated receptor	Phase I clinical trials		Phase II RCT in cardiac surgery (NCT03941483)
Nicotinamide riboside and pterostilbene	Phase I clinical trials		Phase II RCT in complex aortic surgery (NCT04342975)
Hepatocyte growth factor (HGF)-like peptide (ANG-3777)			Phase II RCT in cardiac surgery (NCT02771509)
THR-184 ^a	Phase II RCT	No effect on kidney function	
Mesenchymal stem cells	Phase I clinical trial	Decreased incidence of AKI	Further studies required

^aA bone morphogenetic protein-7 agonist.

AKI, Acute kidney injury; CI-AKI, contrast-induced acute kidney injury; KDIGO, Kidney Disease: Improving Global Outcomes; RCT, randomized controlled trial; RNA, ribonucleic acid.

An RCT including 94 patients undergoing high-risk cardiac surgery showed that prophylactic nesiritide (β -type natriuretic peptide) did not reduce the KRT requirement or lengths of stay, although AKI rates were lower with nesiritide. Conversely, a Japanese RCT including 303 patients with CKD who underwent coronary artery bypass graft (CABG) surgery showed that human atrial natriuretic peptide (hANP, carperitide) decreased postoperative serum creatinine and need for dialysis,²⁶ although other studies using the same medication have shown increased mortality in patients with acute heart failure. Further studies are required.

Vasoactive Agents: Vasopressors, Inotropes, and Vasodilators

Kidney-dose dopamine 0.5 to 3 μ g/kg/min given as a kidney vasodilator increases urine output, but several studies have confirmed that this drug does not affect AKI outcome or mortality.²⁷ Dopexamine, a synthetic dopamine analog, is a dopamine type-1 and less potent dopamine type-2 receptor agonist. Small studies performed in patients undergoing liver transplant surgery have not found a beneficial effect of dopexamine in preventing AKI.

Fenoldopam is a pure dopamine type-1 receptor agonist with hemodynamic kidney effects similar to those of low-dose dopamine,

without systemic α - or β -adrenergic stimulation. In meta-analyses, fenoldopam was shown to reduce the risk for AKI in postoperative or critically ill patients (OR, 0.43–0.46).^{28,29} Intrakidney administration of fenoldopam allows the use of a substantial dose of fenoldopam mesylate while avoiding systemic adverse effects, such as hypotension. Data from experimental models suggest that fenoldopam may have additional anti-inflammatory effects. We do not recommend use of fenoldopam to prevent AKI because no high-quality data support use of this agent.

A meta-analysis on the use of levosimendan, a calcium-sensitizing agent with inotropic and vasodilatory effects, has demonstrated reduced AKI after cardiac surgery, KRT requirement, and mortality. However, a large RCT showed that levosimendan did not reduce mortality or need for KRT.³⁰

Calcium channel blockers (CCBs) have been shown to reverse the afferent arteriolar vasoconstriction induced by a variety of stimuli and also have an independent natriuretic effect. These drugs have been evaluated in the prevention of delayed graft function (DGF). A large multicenter RCT did not find any benefit on the incidence and severity of DGF. A systematic review did not find strong evidence that CCBs reduce the incidence of DGF after transplantation.

Theophylline, a nonselective adenosine receptor antagonist, prevents adenosine-mediated vasoconstriction of the afferent arteriole. Adenosine is released in response to increased luminal chloride concentrations in the distal tubules as part of the tubuloglomerular feedback. Meta-analyses have found a significant reduction in CI-AKI with theophylline.³¹ However, there is insufficient evidence to recommend theophylline monotherapy, and the KDIGO AKI guideline does not suggest using theophylline to prevent CI-AKI.

Selective adenosine blocking agents, such as rolofylline, have been used in trials for prevention and treatment of cardiorenal syndrome. In a small double-blind RCT in decompensated heart failure with AKI, the coadministration of adenosine A_1 antagonist with furosemide increased diuresis and prevented further decrease in GFR.

Statins

Although the pathogenesis of CI-AKI is not completely known, multiple mechanisms may be involved. Statins induce downregulation of angiotensin receptors, decrease endothelin synthesis, decrease inflammation, improve endothelial function by inhibiting NF- κ B, decrease expression of endothelial adhesion molecules, increase NO bioavailability, attenuate production of reactive oxygen species, and protect against complement-mediated injury. Those mechanisms may be involved in CI-AKI. A number of observational studies and some but not all RCTs support the potential for kidney protection with statin before an intra-arterial administration of iodinated contrast. The statins most commonly studied in this setting were rosuvastatin and atorvastatin. Therapy should continue in patients who are already receiving a statin, and those who need statins for another indication such as myocardial infarction may receive statins before their angiography. It is currently unclear whether statins should be specifically initiated to prevent CI-AKI. A recent meta-analysis suggested that CKD patients may benefit more from statins in this setting.³² Regardless, according to the KDIGO lipid guideline, CKD patients aged 50 or older with eGFR less than 60 mL/min/1.73 m² should be treated with a statin. A large recent retrospective study failed to show any benefit of statins for preventing AKI related to administration of IV iodinated contrast.³³

Statins were associated with lower risk of AKI after elective surgery in observational studies. However, RCTs did not confirm these findings. In the largest RCT, the incidence of AKI after cardiac surgery was similar between atorvastatin and placebo groups among all

participants (20.8% vs. 19.5%; risk ratio [RR], 1.06; 95% CI, 0.78–1.46; $P = .75$).³⁴ Similar results were observed when stratifying for statin-naïve patients or statin users.

Antiinflammatory Agents

An alpha-melanocyte-stimulating hormone analog, ABT-719, inhibits inflammatory, cytotoxic, and apoptotic pathways. In a phase IIB study, patients with CKD who underwent cardiac surgery did not have a lower incidence of AKI compared with placebo.

Dexmedetomidine, an alpha-2 adrenergic agonist used for its sedative properties, also has hemodynamic and antiinflammatory effects. A recent meta-analysis in cardiac surgery found that dexmedetomidine might reduce the incidence of AKI.³⁵ A small RCT after aortic surgery has also shown a decreased incidence of AKI.³⁶

Antioxidants

The endogenous antioxidative enzyme heme oxygenase-1 (HO-1) is a stress-inducible enzyme and has important antiapoptotic and anti-inflammatory functions. Preliminary data suggest that the induction of HO-1 may be protective in several forms of injury, including AKI. A phase III RCT including 600 patients is currently determining whether this drug reduces the incidence of DGF after kidney transplantation (HOT 2 Trial, NCT03646344).

Propofol, which has antioxidant and anti-inflammatory properties, can also increase levels of bone morphogenetic protein-7 (BMP-7), a potential tubular repair agent.³⁷ Propofol has previously shown conflicting results on kidney function. There are ongoing RCTs assessing the effect of propofol on kidney function, including one in kidney transplantation (VAPOR-2 Trial, NCT02727296), one after nephrectomy (NCT04474600), and another one in orthopedic surgery patients (NCT03336801).

Apoptosis p53 Inhibitor

In a phase II trial, patients treated with small interfering ribonucleic acid targeting p53 had a reduced AKI rate compared with those treated with placebo. Because p53 has, among other activities, a tumor suppression function, one of the major drawbacks to the use of an inhibitor of p53 is its potential carcinogenic effect. A phase III trial is underway to assess the efficacy of teprasiran to prevent AKI in cardiac surgery (NCT03510897).

Targeting Mitochondrial Stress and Cell Metabolism

ASP1128, a peroxisome proliferator-activated receptor (PPAR) delta modulator, is currently studied in a phase II trial in patients undergoing cardiac surgery (NCT03941483). A phase II trial is ongoing targeting metabolic pathways related to cellular protection using NAD⁺ supplementation with nicotinamide riboside and pterostilbene to prevent AKI in patients undergoing complex aortic surgery (NCT04342975).

Tubular Repair Agents

In a recent phase II RCT in patients at risk for AKI undergoing cardiac surgery, THR-184, a peptide that activates the bone morphogenetic protein pathway, did not reduce AKI over the postoperative week.³⁸ ANG-3777, a hepatocyte growth factor (HGF)-like peptide, is evaluated in a phase II trial in patients having cardiac surgery.

Mesenchymal stem cells (MSCs) were shown to prevent ischemia/reperfusion-induced AKI in rats. A phase I clinical trial evaluated the feasibility and safety of suprakidney aorta infusion of allogeneic MSCs in patients undergoing on-pump cardiac surgery.³⁹ No adverse events were associated with the MSC infusion, and lengths of hospital stay and readmission rates were decreased by 40% compared with matched historical controls. Postoperative kidney function remained at baseline

levels and no patients in the treatment group required hemodialysis (HD), whereas 20% of controls developed AKI. In addition, kidney function in the treatment group was stable for up to 16 months in patients with CKD, whereas matched controls showed progressive deterioration in kidney function.³⁹ The long-term safety of MSC is unknown.

Sodium-Glucose Transporter Inhibitors

Sodium-glucose transporter (SGLT2) inhibitors have shown beneficial cardiovascular and kidney effects in patients with diabetic nephropathy. However, after approval, there were several reports of severe AKI. In recent RCTs and meta-analysis, SGLT2 inhibitors reduced the incidence of AKI by 25%, without heterogeneity across studies. These findings were replicated in real-world analyses. The reasons why these drugs may reduce AKI remain to be determined, and there is no current recommendation to use them to reduce the risk of AKI. Importantly, patients should be instructed to temporarily withhold these agents during sick days.

Summary

To prevent AKI, hypovolemia, hypotension, and sepsis should be quickly addressed. Isotonic fluids and vasopressors can be administered depending on the clinical scenario. Preference should be given to balanced solutions, especially if high volumes will be given in patients at high risk for AKI. HES and other nephrotoxins should definitively be avoided. For prevention of CI-AKI (see Fig. 74.2), patients with eGFR less than 45 mL/min/1.73 m² receiving IV contrast or those with eGFR less than 60 mL/min/1.73 m² receiving intra-arterial contrast should receive IV hydration with crystalloid solution. The benefit of statins and RIPC is controversial. To prevent TLS, patients at intermediate and high risk should receive aggressive hydration with crystalloid solutions. A xanthine oxidase inhibitor or rasburicase should be administered depending on the underlying risk and uric acid levels. For rhabdomyolysis, normal saline is preferable. Additional studies are required to assess whether fenoldopam, certain vasopressors, nesiritide, and statins are beneficial to prevent AKI.

TREATMENT OF ACUTE KIDNEY INJURY

Once measures to prevent AKI have failed, a key question is whether AKI can be managed with nondialytic therapy alone or if KRT is necessary (see Chapter 75). Management of AKI in liver failure is discussed in Chapter 76.

General Management

Appropriate management requires timely diagnosis of the clinical condition. Considerable effort and investment have been directed in the search for a more sensitive and specific biomarker to diagnose AKI. Initial management of AKI includes assessment of the cause and volume status. Adequate hemodynamic status should be maintained to ensure kidney perfusion and avoid further kidney injury. In the injured kidney, autoregulation of blood flow, the mechanism responsible for maintaining a constant flow during fluctuations in BP, is lost. This loss increases the susceptibility to develop AKI after episodes of hypotension. Therefore, fluid and vasoactive drug management is needed in both patients at risk for AKI and with AKI. In patients with AKI attributed to dehydration only, it is expected that the serum creatinine decreases after the administration of 1 to 3 L of isotonic fluid over 24 to 48 hours, depending on the severity of the underlying condition. The optimal MAP is unknown and probably needs to be individualized according to age, hypertension status, and degree of peripheral artery and renovascular disease. Any potentially nephrotoxic agents should

be avoided, including intravascular iodinated contrast. Gadolinium-based contrast agents are associated with the risk of developing nephrogenic systemic fibrosis (NSF) in patients with eGFR lower than 30 mL/min. The risk of NSF is higher with group I (linear) agents and lower with newer group II (cyclic) agents, such as gadobenate dimeglumine, gadobutrol, gadoteridol, and gadoterate meglumine. The risk with group III agents remains unknown. If gadolinium-based contrast agents need to be used in AKI, patients should be informed about the risk for nephrogenic systemic fibrosis, and group II agents should be preferred. The lowest dosage possible should be administered and repeated exposures avoided. Antimicrobial agents such as aminoglycosides, amphotericin, acyclovir, and pentamidine should be avoided whenever possible, or their dose should be adjusted to prevent further insult. Any medications associated with AKI also should be avoided if possible.

Fluid and Electrolyte Management

Although early and vigorous resuscitation with crystalloid solutions and aggressive infection control can reduce the incidence of AKI (see earlier discussion), the role of fluid resuscitation in established AKI not attributed to dehydration is less clear. Volume status is one of the most difficult parameters to assess, and fluid resuscitation should target a predefined preload, SV, or cardiac output rather than a set MAP. Nevertheless, many clinical studies have emphasized the limitations of static measures such as central venous pressure, right atrial pressure, and pulmonary artery occlusion pressure in predicting volume expansion efficacy. Other bedside indicators of preload, such as the right ventricular end-diastolic volume (evaluated by thermodilution) and the left ventricular end-diastolic area (measured by echocardiography), are ineffective in differentiating volume responder from nonresponder patients. As previously discussed, two techniques have an acceptable degree of clinical accuracy to determine fluid responsiveness: the passive leg raising maneuver and the fluid bolus test both coupled with real-time SV monitoring. In ICU patients receiving mechanical ventilation, respiratory changes in left ventricular SV also can predict fluid responsiveness. In hypovolemic patients, positive-pressure ventilation may induce a fall in the venous return and consequently in cardiac output. Based on the positive relationship between ventricular end-diastolic volume and SV, the expected hemodynamic response to volume expansion is an increase in right ventricular end-diastolic volume, left ventricular end-diastolic volume, SV, and cardiac output.

Volume expansion in ICU patients can frequently result in a relative increase in body weight of 10% to 15% or more, sometimes doubling the total body water in a short time. As previously highlighted, studies have demonstrated an association between fluid accumulation and mortality in AKI. A prospective multicenter observational study found that fluid overload at AKI diagnosis (defined as an increase in body weight of $\geq 10\%$ relative to baseline) was independently associated with increased mortality.⁷ The risk for death was proportional to the magnitude and duration of fluid accumulation. However, the effect of fluid overload on kidney recovery was inconsistent. A second analysis from the Fluids and Catheters Treatment Trial (FACTT) in AKI patients confirmed that in early AKI, positive fluid balance is strongly associated with mortality. The study showed a protective effect of furosemide on mortality, which disappeared after adjustment for fluid balance. Other studies have shown a deleterious effect of fluid overload on kidney function. In summary, results from observational studies suggest that a conservative fluid approach may be beneficial in terms of mortality and kidney recovery in patients with severe AKI. Fluid resuscitation should be performed rapidly if there is hypoperfusion and then ideally should be guided based on the aforementioned dynamic measures. After the initial resuscitation period, treatment should focus

on preventing further fluid overload, monitoring signs of fluid overload, and ensuring early removal of excess volume. However, RCTs are required to confirm these findings before any formal recommendation can be made. Of note, in the STARRT-AKI RCT comparing standard versus accelerated strategies for KRT initiation, even in the subgroup where fluid overload was more common in the standard compared with the accelerated group (6.7% of body weight vs. 3.1%), 90-day mortality was similar between groups.

Drugs to Promote Recovery From Acute Kidney Injury

See Table 74.2 for a summary of drugs used in treatment of AKI.

Loop Diuretics

Although loop diuretics are often prescribed in established AKI, a meta-analysis confirmed that they are not associated with reduced mortality or improved kidney recovery.⁴⁰ Other meta-analyses have shown that loop diuretics do not affect mortality, need for dialysis, or the number of dialysis sessions. Concomitant prescription of aminoglycosides and diuretics should be avoided because of an increased risk of ototoxicity. We suggest using diuretics to manage fluid overload as needed but not in attempts to speed recovery from AKI per se.

Natriuretics

Atrial natriuretic peptide (ANP) has been studied as a treatment for AKI in four RCTs. ANP was initially shown to reduce need for dialysis but not mortality. In the largest study published, ANP improved

dialysis-free survival in the subgroup of oliguric patients only. Unfortunately, a subsequent trial including 222 oliguric patients did not confirm that ANP reduces mortality or dialysis-free survival.⁴¹ A smaller, more recent study showed that ANP decreased the probability of dialysis and improved dialysis-free survival. Therefore, larger studies are required to confirm the benefits of ANP in AKI. Nesiritide, a β -type natriuretic peptide, has been studied to treat heart failure. Nesiritide induces vasodilation and an indirect increase in cardiac output but has no inotropic effects and neutral effect on heart rate. In addition, it inhibits adverse neurohormonal activation and can result in natriuresis and diuresis in some individuals. However, in a large RCT in patients with acute heart failure, this drug did not decrease mortality or rehospitalization rates and had a nonsignificant effect on dyspnea.⁴² Nesiritide did not adversely affect kidney function, but it increased hypotension. Nesiritide also has been assessed in high-risk cardiovascular surgery, in which it reduced AKI rates in the immediate postoperative period but did not improve long-term survival. The KDIGO guideline does not support the use of ANP or nesiritide to treat AKI.

Vasoactive Agents: Vasopressors, Inotropes, and Vasodilators

Dopamine use to treat AKI is not recommended (see earlier discussion). Vasopressors often have been considered detrimental for organ perfusion. In septic shock, a small prospective study showed that norepinephrine improved serum creatinine and creatinine clearance when MAP was raised above 70 mm Hg. However, in another small RCT, increasing MAP from 65 to 85 mm Hg with norepinephrine did not

TABLE 74.2 Drugs Used in Treatment of Acute Kidney Injury

Drug	Level of Evidence	Results	Comments
Dopamine	RCTs	No effect on mortality or kidney function	
Fenoldopam	Small RCTs One meta-analysis	No effect on mortality or kidney function Beneficial effect on mortality and need for dialysis	Further studies required
Norepinephrine	Prospective observational studies	Possible beneficial effect on kidney function	Further studies required
Angiotensin II	Post hoc analysis from one RCT including patients with vasodilatory shock on dialysis	Decreased dialysis-dependence at 7 days	Further studies required; one RCT in hepatorenal syndrome (NCT04048707)
Loop diuretics	RCTs and meta-analyses	No effect on kidney function	
Atrial natriuretic peptide	RCTs	Possible beneficial effect on survival and kidney function	Further studies required
B-type natriuretic peptide	RCT in acute heart failure	No effect on kidney function	
Levosimendan	Meta-analysis in critically ill patients at risk or with AKI	Decreased need for dialysis	Ongoing phase 4 trial in cardiac surgery (LEVOAKI, NCT02531724)
Mesenchymal stem cells	Animal models and human studies	Beneficial effect on kidney function in animal models but no effect in one human RCT	Ongoing phase II studies (NCT04194671, NCT03015623). Study NCT03015623 showed that mesenchymal stem cells can trigger a phenotypic switch from tissue injury to repair, but the study was not powered for clinical efficacy.
Alkaline phosphatase	Small phase II RCT	No effect on kidney function in sepsis	Further studies required; ongoing phase III trial (NCT04411472)
CD28 receptor antagonist (AB 103 – Reltecimod)	Phase III RCT in necrotizing soft tissue infections	Improved organ dysfunction	Further studies required; ongoing phase II trial in peritonitis or necrotizing soft tissue infections (NCT03403751)
Hepatocyte growth factor (HGF)-like peptide (ANG-3777)			Phase III trial in kidney transplants with delayed graft function (NCT02474667)

AKI, Acute kidney injury; RCT, randomized controlled trial.

improve kidney function. No studies have compared the effect of different target MAPs to treat AKI.

A posthoc analysis of the ATHOS-3 data found that septic patients with AKI requiring dialysis who received angiotensin II were more likely to be dialysis-free at 7 days compared with placebo.⁴³ Confirmation of these results in large prospective studies is needed.

In a meta-analysis, fenoldopam decreased the need for dialysis (7% vs. 10%) and in-hospital mortality (15% vs. 19%) in postoperative or ICU patients.²⁸ Several limitations were present in this meta-analysis, such as absence of an independent measure of GFR and standardized criteria for initiation of dialysis, heterogeneity of populations, AKI definitions, dosage, and duration of treatments. In addition, fenoldopam has hypotensive properties and may be more dangerous in real-world patients than in selected trial participants. No individual trial has shown that fenoldopam can reduce the need for dialysis. The results of the meta-analysis should be confirmed with an adequately powered trial, and like the KDIGO guideline, we do not suggest using fenoldopam to treat AKI.

One small RCT is currently assessing the effect of levosimendan on kidney outcomes after cardiac surgery in patients with AKI (LEVOAKI, NCT02531724).

Specific therapy for patients with the hepatorenal syndrome includes the use of terlipressin in combination with octreotide (see [Chapter 76](#)). In countries such as the United States, where terlipressin is not available, a combination of midodrine, octreotide, and albumin infusions is often used. Norepinephrine also has been used in these settings with good response equivalent to that of terlipressin. One trial will test the efficacy of angiotensin II in hepatorenal syndrome (ANTHEM Trial, NCT04048707).

Tubular Repair Agents

Other agents have been studied for treatment of established AKI. ANG-3777, a hepatocyte growth factor (HGF)-like peptide, is evaluated in a phase III trial in kidney transplants having delayed graft function aiming to activate HGF-related repair pathways. One promising therapy is MSCs. MSCs are multipotent cells with anti-inflammatory and immunomodulatory properties proven to be beneficial in animal models of myocardial ischemia, sepsis, and AKI. In animal models, infusion of MSCs improved recovery of kidney function in cisplatin-induced ischemia/reperfusion injury and glycerol-induced AKI. A dose-escalating phase I clinical trial was conducted to test the safety and preliminary efficacy of MSCs in patients at high risk for AKI.³⁹ An experimental study using a model of progressive mesangioproliferative nephritis evaluated the long-term effects of intrakidney, syngeneic MSC transplantation. Although rats in the MSC-treated group had lower proteinuria and better kidney function on day 60, 20% of the glomeruli of MSC-treated rats contained single or clusters of large adipocytes with pronounced surrounding fibrosis. Therefore, the MSC benefit of maintaining kidney function in the short term needs to be balanced with a possible long-term effect of partial maldifferentiation of intraglomerular MSC into adipocytes and subsequent glomerular sclerosis. There are ongoing studies on the use of MSC to improve AKI outcomes (NCT04194671, NCT03015623). However, in a phase II trial of patients with cardiac surgery, no benefit from MSCs, administered into the aorta at the time of surgery, could be demonstrated.^{43a}

Antiinflammatory Agents

In severe sepsis and septic shock, a small study has shown that the infusion of alkaline phosphatase, an enzyme expressed along the brush border of the proximal tubule, improves kidney function,

possibly through reduced kidney inflammation. A phase II study assessing the effect of recombinant alkaline phosphatase in septic patients with AKI failed to show an improvement in 7-day kidney function.⁴⁴ However, there were dose-dependent signals of beneficial effects on kidney function and mortality, prompting a phase III RCT (NCT04411472).

In a phase III RCT including critically ill patients with necrotizing soft tissue infections, reltecimod, a CD28 receptor antagonist, improved resolution of organ dysfunction/failure by day 14.⁴⁵ This agent was designed to help control the cytokine storm, by modulating rather than inhibiting acute inflammation. A phase II including patients with peritonitis or necrotizing soft tissue infections is currently underway to determine whether this drug can improve kidney recovery (NCT03403751).

Summary

Dehydration, hypotension, and infections should be rapidly treated. Any potentially nephrotoxic agents should be avoided, including intravascular iodinated contrast. Gadolinium-based contrast agents should also be avoided. Fluid overload should be minimized and eventually corrected once the patient's clinical condition permits. No agent has been clearly shown to facilitate kidney recovery. Several studies are ongoing to test the efficacy of new drugs to treat AKI in specific conditions.

TREATMENT OF ACUTE KIDNEY INJURY COMPLICATIONS

Fluid Overload

When fluid overload occurs in AKI, all intakes should be minimized and medical treatment should be attempted before dialysis initiation. In patients with positive fluid balance with large fluid intake and inadequate urine output and in those presenting with symptomatic volume overload, a loop diuretic can be initiated with measures to optimize systemic and kidney perfusion. IV bolus doses of diuretics may be necessary to optimize the response, especially in patients with heart failure and nephrotic syndrome. If there is a response to an IV bolus, continuous infusion can be considered because it is less ototoxic. When starting or increasing the dose of a continuous infusion, a bolus should be administered before. Natriuretic peptides inhibit sodium reabsorption in the nephron, resulting in net sodium excretion. There is currently no evidence to support the use of natriuretic peptides as an adjunctive treatment in AKI.

Morphine and nitrates can be used to alleviate the respiratory symptoms of pulmonary edema in urgent situations. Morphine can be administered intravenously at an initial dose of 2 to 4 mg over a 3-minute period and repeated at 5- to 15-minute intervals as needed. Nitrates are also commonly used. Nitroglycerin reduces left ventricular filling pressure through venodilation; an initial dose of 5 µg of IV nitroglycerin per minute can be used. When fluid overload cannot be rapidly treated with medical management, positive-pressure ventilation may be initiated with or without endotracheal intubation and dialysis, depending on the clinical situation.

Potassium Disorders

Hyperkalemia, covered in detail in [Chapter 10](#), is a frequent complication of AKI that may affect cardiac conduction and lead to bradycardia or even asystole. If electrocardiographic changes are present, IV administration of calcium is urgently needed. Sources of potassium should be removed, including drugs with effect on potassium handling, such as

β -adrenergic antagonists, potassium-sparing diuretics, ACE inhibitors, and ARBs.

The next step is to enhance the shift of potassium to the intracellular space by parenteral glucose and insulin infusions. Sodium bicarbonate also promotes shift of potassium and is most efficient when there is concomitant metabolic acidosis. It can be started if there is no concern regarding fluid overload. β -Adrenergic agonists given as aerosols are effective at very high doses but more likely to produce side effects and therefore are rarely prescribed to treat hyperkalemia.

Potassium excretion should be increased by the administration of saline, loop diuretics, and cation exchange resins such as sodium polystyrene sulfonate or calcium resins. The resins can be administered orally or rectally as a retention enema and are usually not effective after a single dose only. There is concern that coadministration of sorbitol with sodium polystyrene sulfonate may promote intestinal necrosis. Sodium polystyrene sulfonate should be avoided postoperatively until normal bowel function resumes and in patients with bowel obstruction or ileus. Newer agents, such as sodium zirconium cyclosilicate (SZC) and patiomer, have shown reductions in potassium levels within hours of administration. SZC has been effective in normalizing average potassium levels of 5.5 mmol/L (range, 5.1–7.3) in 82% of subjects within 24 hours, and in 96% within 48 hours.⁴⁶ If hyperkalemia is unresponsive to conservative measures, emergency HD is the treatment of choice. If HD is unavailable, continuous KRT can be used with higher volumes of replacement solution and/or dialysate and low or no potassium content. Because KRT initiation usually takes a few hours, medical management always should be used pending initiation of dialysis. Monitoring of potassium levels should continue after conservative or dialytic management to prevent and treat rebound hyperkalemia.

Sodium Disorders

Hyponatremia can occur as a result of relative impairment in free water excretion and is more common in AKI associated with heart failure, liver failure, or diuretics. In these settings, water restriction to below the level of output is mandatory. Sodium restriction is usually necessary to treat fluid overload and edema. In patients with true volume depletion and prerenal AKI, isotonic saline needs to be administered to correct both disorders.

In patients with AKI and hypernatremia, treatment of the cause is necessary and water deficit should be estimated. Water should be administered orally or intravenously as dextrose in water to correct serum sodium concentration at a maximum rate of 12 mmol/L/day. Dialysis and continuous KRT may be required to optimally correct sodium disorders in AKI.

Calcium, Phosphorus, and Magnesium Disorders

Hyperphosphatemia and hypocalcemia are common in AKI. Hyperphosphatemia is usually caused by reduced excretion by the kidneys, and continuous release from rhabdomyolysis or TLS also can be contributive (see [Chapters 11 and 70](#)). As phosphorus levels increase, calcium levels decrease, resulting in hypocalcemia. Total calcium levels usually drop to 7 to 8 mg/dL (1.75–2.0 mmol/L). Other causes of hypocalcemia in AKI are skeletal resistance to parathyroid hormone (PTH) and low calcitriol production. Hypocalcemia is aggravated when bicarbonate is administered to correct acidosis. A high-calcium, high-phosphorus product could theoretically trigger tissue calcium deposition, which can cause cardiac arrhythmia. No RCT has evaluated the benefits of treating these disorders. However, because hyperphosphatemia caused by oral phosphorus-containing medications and TLS can cause AKI,

severe hyperphosphatemia should be avoided to prevent further damage. Calcium-based phosphate binders and other phosphate binders can be used in this setting. If there is symptomatic hypocalcemia or hemodynamic instability, calcium gluconate infusion should be administered.

Hypercalcemia is rare in AKI and is usually seen in the recovery phase of rhabdomyolysis when calcium is released from calcium-containing complexes in muscle (see [Chapters 11 and 70](#)). In addition, when production of calcitriol is reestablished by the recovering kidney, an enhanced responsiveness to PTH can be seen. Hypercalcemia in this setting is rarely problematic and can be easily treated with medical management. Mild hypermagnesemia is frequent in AKI and usually does not have clinical consequences.

Acid-Base Disorders

In AKI, normal anion gap metabolic acidosis is the most common acid-base abnormality (see [Chapter 13](#)) because of reduced regeneration of bicarbonate and failure to excrete ammonium ions. Accumulation of phosphate and unexcreted unmeasured anions, such as sulfate, urate, hippurate, hydroxypropionate, furanpropionate, and oxalate, is contributory. Hypoalbuminemia can attenuate this acidification process, and it is exacerbated by lactic acidosis. Despite retention of unmeasured anions, the anion gap remains within normal in 50% of patients. Although metabolic acidosis is frequent, triple acid-base disturbances also can occur. The approach to acid-base disturbances in AKI needs to be adjusted according to the underlying causes.

When metabolic acidosis is a complication of AKI, sodium bicarbonate can be administered if the serum bicarbonate concentrations fall below 15 to 18 mmol/L. In stable patients, oral sodium bicarbonate may be given because volume overload can occur after the administration of IV bicarbonate. The administration of bicarbonate in shock with lactic acidosis is controversial, given the possibility of an increase in carbon dioxide generation, worsening of the intracellular acidosis, and volume overload. In addition, bicarbonate therapy may reduce calcium and potassium levels. However, the recent BICAR-ICU trial has shown that in critically ill patients with AKI stage 2 or higher and lactic acidosis with pH less than 7.2, bicarbonate administration may prevent the need for dialysis and improve survival.⁴⁷ Alternative forms of base treatment such as tris(hydroxymethyl)aminomethane (THAM) are not recommended in patients with AKI because THAM can cause hyperkalemia. Restriction of protein intake also has been suggested as a method of acidosis control because protein breakdown has been associated with worsening acidosis, as in CKD. However, protein restriction is not recommended in AKI.

Nutrition

Patients with AKI have an increased risk for protein-energy malnutrition because of poor nutrient intake and high catabolic rate. Nutritional assessment is difficult, especially in AKI patients with high metabolic demands. Subjective global assessment evaluates nutritional status, requires no additional laboratory testing, and is highly predictive of outcome.

Patients with AKI should receive a basic intake of 0.8 to 1.0 g of protein/kg/day if not catabolic and a total energy intake of 20 to 30 kcal/kg/day, as recommended in the KDIGO guidelines. In addition, in patients on KRT, 1.0 to 1.5 g of protein/kg/day should be administered up to 1.7 g/kg/day in continuous KRT and hypercatabolic patients. The enteral route should be favored if the gastrointestinal tract is functioning; parenteral nutrition should be prescribed only when the gastrointestinal tract cannot be used or when the enteral route is inadequate to reach nutrient intake goals.

SELF-ASSESSMENT QUESTIONS

1. In a general population of critically ill patients, administering starches has proven to be:
 - A. beneficial for kidney function (decrease in need for KRT).
 - B. beneficial for hypotension.
 - C. detrimental for kidney function (increase in need for KRT).
 - D. detrimental for hypotension.
2. Which of the following drugs should be stopped, if feasible, 48 hours before iodinated contrast administration in patients with chronic kidney disease and a glomerular filtration rate below 45 mL/min/1.73 m²?
 - A. Furosemide
 - B. Metformin
 - C. NSAIDs
 - D. Thiazide diuretics
 - E. All the above
3. The measures to prevent AKI in patients with traumatic rhabdomyolysis include all of these statements *except*:
 - A. Urine output should be maintained around 300 mL/h until CK levels are lower than 1000 U/L.
 - B. Urine output should be maintained around 300 mL/h until myoglobinuria disappears.
 - C. Mannitol may be beneficial because of diuretic, antioxidant, and vasodilatory properties.
 - D. Mannitol can prevent kidney tubular cast deposition, expand extracellular volume, and reduce intracompartmental pressure, muscle edema, and pain.
 - E. Monitoring of ionized calcium is important to avoid NaHCO₃-induced hypocalcemia (as a result of metabolic alkalosis).
4. Regarding fluid administration for critically ill patients at risk for AKI, all the following statements are true *except*:
 - A. Chloride-restrictive fluids (lactated solution, Plasma-Lyte 148, or chloride-poor 20% albumin) seem preferable to chloride-rich intravenous fluids (0.9% saline, 4% succinylated gelatin solution, or 4% albumin solution), although further studies are needed and ongoing.
 - B. Volume expansion should be maintained after major vascular surgeries unless patients develop evidence of pulmonary congestion.
 - C. Guidelines on sepsis management mention that septic patients should receive initial fluid resuscitation with crystalloids at a minimum of 30 mL/kg.
 - D. Intravenous administration of albumin can be considered in patients with sepsis who require substantial amounts of crystalloid to maintain adequate mean arterial pressure.
 - E. Fluid expansion should be stopped when patients are no longer fluid responsive.

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